

Question	Answer	Marks
2(a)(i)	straight line through origin indicates acceleration $\propto$ displacement	B1
	negative gradient shows acceleration and displacement are in opposite directions	B1
2(a)(ii)	$a = -\omega^2 y$ and $\omega = 2\pi f$	C1
	$4.5 = (2\pi \times f)^2 \times 8.0 \times 10^{-3}$ (or other valid read-off)	
	$f = 3.8$ Hz	A1
2(b)(i)	maximum displacement upwards/above rest/above the equilibrium position	B1
2(b)(ii)	(just leaves plate when) acceleration = 9.81 ms^{-2}	C1
	$9.81 = (2\pi \times 3.8)^2 \times y_0$ or $9.81 = 563 \times y_0$	C1
	amplitude = 17 mm	A1

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2(a)(i)	sum of potential and kinetic energies (of molecules/atoms/particles)	D1